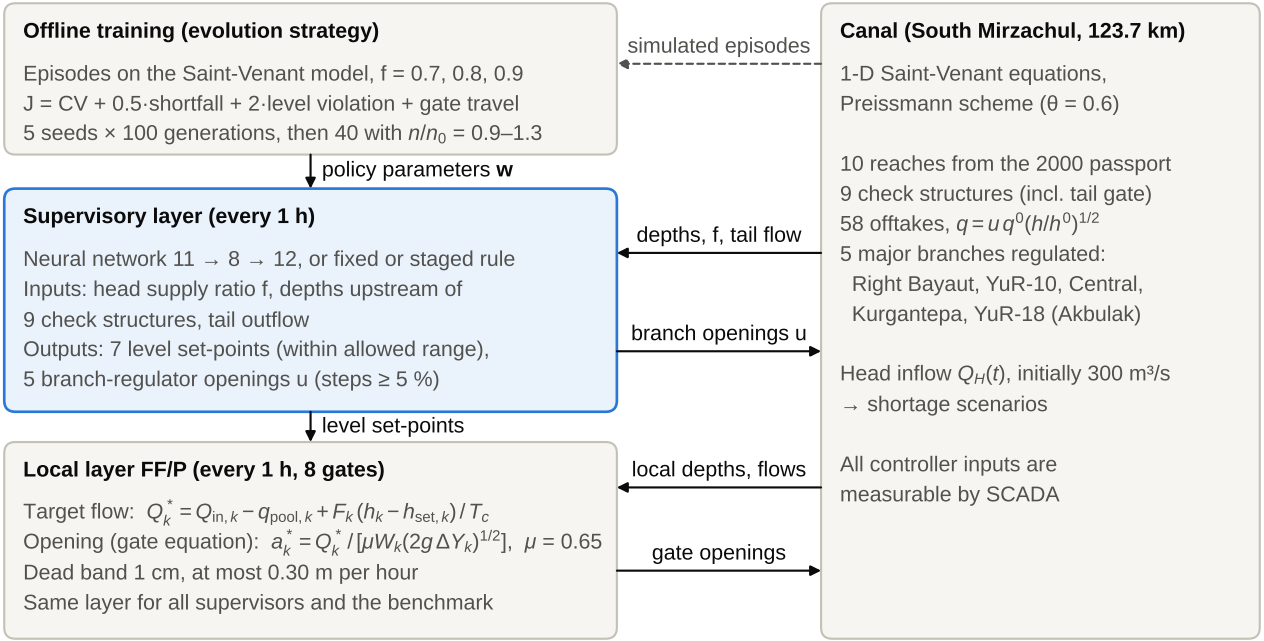
Offline training (evolution strategy)
Episodes on the Saint-Venant model, f = 0.7, 0.8, 0.9
J = CV + 0.5·shortfall + 2·level violation + gate travel
5 seeds × 100 generations, then 40 with n/n0 = 0.9–1.3
policy parameters w
Supervisory layer (every 1 h)
Neural network 11 → 8 → 12, or fixed or staged rule
Inputs: head supply ratio f, depths upstream of
9 check structures, tail outflow
Outputs: 7 level set-points (within allowed range),
5 branch-regulator openings u (steps ≥ 5 %)
level set-points
Local layer FF/P (every 1 h, 8 gates)
Target flow: Q*k = Qin,k − qpool,k + Fk(hk − hset,k) / Tc
Opening (gate equation): a*k = Q*k / [μWk(2g ΔYk)^1/2], μ = 0.65
Dead band 1 cm, at most 0.30 m per hour
Same layer for all supervisors and the benchmark
simulated episodes
depths, f, tail flow
branch openings u
local depths, flows
gate openings
Canal (South Mirzachul, 123.7 km)
1-D Saint-Venant equations,
Preissmann scheme (θ = 0.6)
10 reaches from the 2000 passport
9 check structures (incl. tail gate)
58 offtakes, q = u q0(h/h0)^1/2
5 major branches regulated:
Right Bayaut, YuR-10, Central,
Kurgantepa, YuR-18 (Akbulak)
Head inflow QH(t), initially 300 m³/s
→ shortage scenarios
All controller inputs are
measurable by SCADA